

Probability (Set B) — MCQ Worksheet

Mathematics • Chapter 15 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Probability of impossible event is:

- | | |
|--------|---------------|
| (A) 0 | (B) 1 |
| (C) -1 | (D) Undefined |

Q2. Probability of sure event is:

- | | |
|---------|---------|
| (A) 0 | (B) 1 |
| (C) 0.5 | (D) 100 |

Q3. Probability of getting a tail on tossing a coin:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{2}$ | (B) 0 |
| (C) 1 | (D) $\frac{1}{4}$ |

Q4. Probability of getting an even number on rolling a die:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{2}$ | (B) $\frac{1}{3}$ |
| (C) $\frac{1}{6}$ | (D) $\frac{2}{3}$ |

Q5. From a deck of 52 cards, probability of drawing a heart:

- | | |
|-------------------|--------------------|
| (A) $\frac{1}{4}$ | (B) $\frac{1}{13}$ |
| (C) $\frac{1}{2}$ | (D) $\frac{1}{26}$ |

Q6. Probability of drawing a queen from a deck:

- | | |
|--------------------|--------------------|
| (A) $\frac{1}{13}$ | (B) $\frac{4}{13}$ |
| (C) $\frac{1}{4}$ | (D) $\frac{1}{52}$ |

Q7. Probability of drawing a black king:

- | | |
|---|--------------------|
| (A) $\frac{1}{26}$ | (B) $\frac{2}{52}$ |
| (C) $\frac{1}{26}$ or $\frac{2}{52}$ (same) | (D) $\frac{1}{13}$ |

Q8. From numbers 1 to 50, probability of choosing a prime:

- | | |
|---------------------|---------------------|
| (A) $\frac{15}{50}$ | (B) $\frac{12}{50}$ |
| (C) $\frac{20}{50}$ | (D) $\frac{10}{50}$ |

Q9. From a bag with 4 red and 6 blue marbles, probability of drawing red:

- | | |
|--------------------|--------------------|
| (A) $\frac{4}{10}$ | (B) $\frac{6}{10}$ |
| (C) $\frac{1}{10}$ | (D) $\frac{4}{6}$ |

Q10. If $P(E) = 0.7$, $P(\text{not } E) =$

- | | |
|---------|---------|
| (A) 0.3 | (B) 0.7 |
| (C) 1.7 | (D) 0 |

Q11. Two coins tossed together. Probability of at least one head:

- | | |
|-------------------|-------------------|
| (A) $\frac{3}{4}$ | (B) $\frac{1}{4}$ |
| (C) $\frac{1}{2}$ | (D) 1 |

Q12. Probability of getting a sum of 7 when two dice are rolled:

- | | |
|--------------------|--------------------|
| (A) $\frac{1}{6}$ | (B) $\frac{1}{9}$ |
| (C) $\frac{5}{36}$ | (D) $\frac{7}{36}$ |

Q13. From a bag of 50 ticket numbered 1 to 50, probability of choosing multiple of 5:

- | | |
|---------------------|-------------------|
| (A) $\frac{10}{50}$ | (B) $\frac{1}{5}$ |
| (C) $\frac{5}{50}$ | (D) Both A and B |

Q14. If a coin is tossed thrice, total outcomes are:

- | | |
|-------|--------|
| (A) 6 | (B) 8 |
| (C) 9 | (D) 12 |

Q15. Probability of getting a face card from a deck:

- (A) $\frac{3}{13}$ (B) $\frac{12}{52}$
(C) Both A and B (same) (D) $\frac{4}{13}$

Q16. Probability of getting a doublet (same number on both dice) when 2 dice are thrown:

- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$
(C) $\frac{1}{12}$ (D) $\frac{1}{36}$

Q17. A box has 3 black, 4 white and 5 red balls. One is drawn. Probability it is not black:

- (A) $\frac{3}{12}$ (B) $\frac{9}{12}$
(C) $\frac{1}{12}$ (D) $\frac{5}{12}$

Q18. From a well-shuffled deck, two cards are drawn at random without replacement. Probability both are aces:

- (A) $\frac{1}{221}$ (B) $\frac{1}{169}$
(C) $\frac{1}{13}$ (D) $\frac{1}{52}$

Q19. A bag contains 15 balls of which x are black. If 5 more black balls are added, probability of drawing black is double. Find x .

- (A) 5 (B) 3
(C) 8 (D) 10

Q20. A jar contains 24 marbles, some green and others blue. If a marble is drawn at random, probability of being green is $\frac{2}{3}$. Number of blue marbles:

- (A) 8 (B) 12
(C) 16 (D) 20

Q21. Two unbiased dice are rolled. Probability that sum is multiple of 3:

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$
(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Q22. Box A has balls numbered 1-10. Box B has 5-20. One ball is drawn from each. Probability that both numbers are even:

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Q23. Probability of selecting a vowel from word 'PROBABILITY':

- (A) $\frac{4}{11}$ (B) $\frac{3}{11}$
(C) $\frac{1}{2}$ (D) $\frac{2}{11}$

Q24. A and B are events such that $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cup B) = 0.5$. Find $P(A \cap B)$:

- (A) 0.2 (B) 0.1
(C) 0.3 (D) 0.4

Q25. Three coins are tossed together. Probability of getting exactly 2 heads:

- (A) $\frac{3}{8}$ (B) $\frac{1}{8}$
(C) $\frac{5}{8}$ (D) $\frac{2}{8}$